

International Data Encryption Algorithm (IDEA)

Overview

- DES algorithm has been a popular secret key encryption algorithm and is used in many commercial and financial applications. However, its key size is too small by current standards and its entire 56 bit key space can be searched in approximately 22 hours
- IDEA is a block cipher designed by Xuejia Lai and James L. Massey in 1991
- It is a minor revision of an earlier cipher, PES (Proposed Encryption Standard)
- IDEA was originally called IPES (Improved PES) and was developed to replace DES

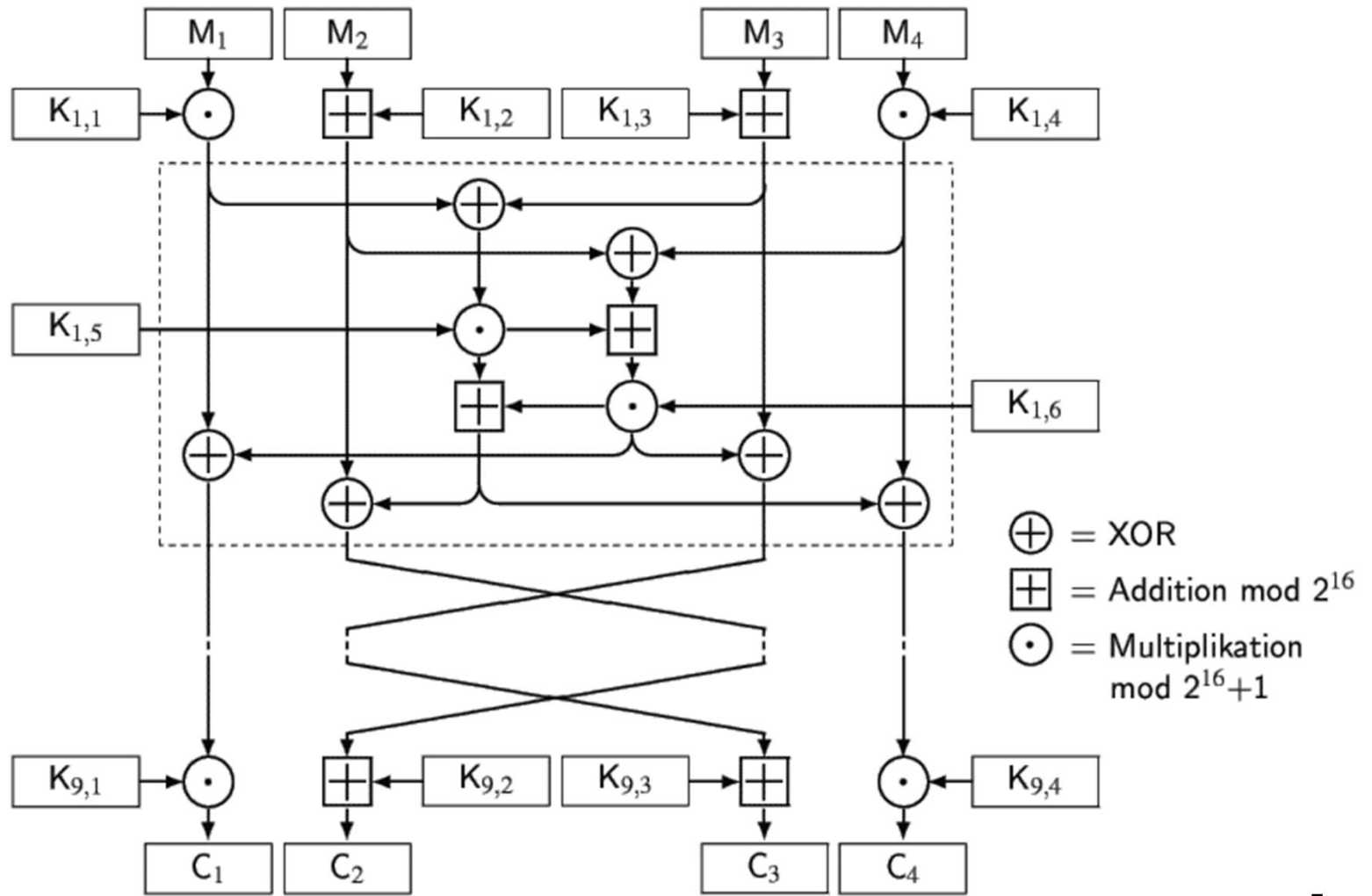
Overview (cont')

- It entirely avoids the use of any lookup tables or S-boxes
- IDEA was used as the symmetric cipher in early versions of the Pretty Good Privacy cryptosystem

Detailed description of IDEA

- IDEA operates with 64-bit plaintext and cipher text blocks and is controlled by a 128-bit key
- Completely avoid substitution boxes and table lookups used in the block ciphers
- The algorithm structure has been chosen such that when different key sub-blocks are used, the encryption process is identical to the decryption process

IDEA – Block Diagram



Key generation

- The 64-bit plaintext block is partitioned into four 16-bit sub-blocks
- Six 16-bit key are generated from the 128-bit key.
- Since a further four 16-bit key-sub-blocks are required for the subsequent output transformation, a total of 52 ($= 8 \times 6 + 4$) different 16-bit sub-blocks have to be generated from the 128-bit key.

Key generation process

- First, the 128-bit key is partitioned into eight 16-bit sub-blocks which are then directly used as the first eight key sub-blocks
- The 128-bit key is then cyclically shifted to the left by 25 positions, after which the resulting 128-bit block is again partitioned into eight 16-bit sub-blocks to be directly used as the next eight key sub-blocks
- The cyclic shift procedure described above is repeated until all of the required 52 16-bit key sub-blocks have been generated

Encryption of the key sub-blocks

- The key sub-blocks used for the encryption and the decryption in the individual rounds are shown in Table 1

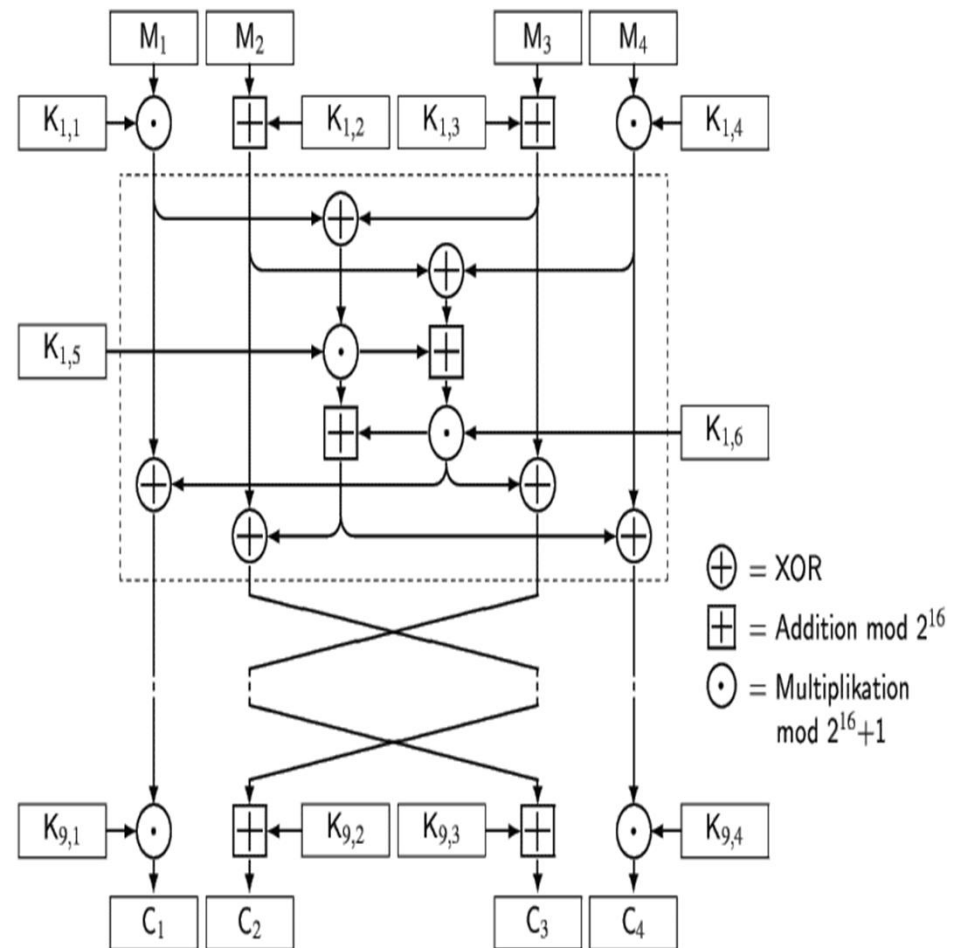
Encryption of the key sub-blocks

Round 1	$Z_1^{(1)} Z_2^{(1)} Z_3^{(1)} Z_4^{(1)} Z_5^{(1)} Z_6^{(1)}$
Round 2	$Z_1^{(2)} Z_2^{(2)} Z_3^{(2)} Z_4^{(2)} Z_5^{(2)} Z_6^{(2)}$
Round 3	$Z_1^{(3)} Z_2^{(3)} Z_3^{(3)} Z_4^{(3)} Z_5^{(3)} Z_6^{(3)}$
Round 4	$Z_1^{(4)} Z_2^{(4)} Z_3^{(4)} Z_4^{(4)} Z_5^{(4)} Z_6^{(4)}$
Round 5	$Z_1^{(5)} Z_2^{(5)} Z_3^{(5)} Z_4^{(5)} Z_5^{(5)} Z_6^{(5)}$
Round 6	$Z_1^{(6)} Z_2^{(6)} Z_3^{(6)} Z_4^{(6)} Z_5^{(6)} Z_6^{(6)}$
Round 7	$Z_1^{(7)} Z_2^{(7)} Z_3^{(7)} Z_4^{(7)} Z_5^{(7)} Z_6^{(7)}$
Round 8	$Z_1^{(8)} Z_2^{(8)} Z_3^{(8)} Z_4^{(8)} Z_5^{(8)} Z_6^{(8)}$
Output Transform	$Z_1^{(9)} Z_2^{(9)} Z_3^{(9)} Z_4^{(9)}$

Table 1

Encryption

- the first four 16-bit key sub-blocks are combined with two of the 16-bit plaintext blocks using addition modulo 2^{16} , and with the other two plaintext blocks using multiplication modulo $2^{16} + 1$
- At the end of the first encryption round four 16-bit values are produced which are used as input to the second encryption round
- The process is repeated in each of the subsequent 7 encryption rounds
- The four 16-bit values produced at the end of the 8th encryption round are combined with the last four of the 52 key sub-blocks using addition modulo 2^{16} and multiplication modulo $2^{16} + 1$ to form the resulting four 16-bit ciphertext blocks



Decryption

- The computational process used for decryption of the ciphertext is essentially the same as that used for encryption
- The only difference is that each of the 52 16-bit key sub-blocks used for decryption is the inverse of the key sub-block used during encryption
- In addition, the key sub-blocks must be used in the reverse order during decryption in order to reverse the encryption process

Applications of IDEA

- Today, there are hundreds of IDEA-based security solutions available in many market areas, ranging from Financial Services, and Broadcasting to Government
- The IDEA algorithm can easily be embedded in any encryption software. Data encryption can be used to protect data transmission and storage. Typical fields are:
 - Audio and video data for cable TV, pay TV, video conferencing, distance learning
 - Sensitive financial and commercial data
 - Email via public networks
 - Smart cards

Conclusion

- As electronic communications grow in importance, there is also an increasing need for data protection
- When PGP (Pretty Good Privacy) was designed, the developers were looking for maximum security. IDEA was their first choice for data encryption
- The fundamental criteria for the development of IDEA were military strength for all security requirements and easy hardware and software implementation